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Do Temperament Trajectories from Late Childhood through Adolescence Predict Success in School? Findings from a Longitudinal Study of Mexican-Origin Youth

Rongxin Cheng[1], **Katherine M. Lawson**[2], **Richard W. Robins**[1]

[1]University of California, Davis

[2]Rhodes College

Abstract

School achievement has long-term consequences for occupational success, mental health, and overall psychological adjustment. The present study examined the association between temperament trajectories from late childhood through adolescence and academic outcomes during late adolescence and young adulthood. Data come from the California Families Project, a longitudinal study of 674 Mexican-origin youth assessed 12 times from age 10 to 23, and from school records. Results from latent growth curve models indicate that higher levels of Effortful Control (EC) at age 10 were associated with better academic achievement (i.e., higher high school GPA and test scores, greater likelihood of high school graduation and college attendance) in late adolescence and young adulthood. Higher levels of Negative Emotionality (NEM) at age 10 were associated with worse academic achievement, but this effect did not hold for all facets of NEM. Neither the levels nor slopes of Positive Emotionality (Surgency, Affiliation) consistently predicted school achievement. There were no main effects of the EC or NEM slopes; however, statistically significant interactions between these slopes and parental monitoring emerged. When parental monitoring was low, youth who experienced greater increases in EC (versus flat or decreasing slopes) had better academic achievement and youth who experienced greater increases in NEM had worse academic achievement; in contrast, when parents closely monitored their children, changes in EC and NEM were only weakly associated with achievement. Overall, these findings demonstrate that temperament in late childhood, and changes in temperament across adolescence, have important prospective effects on academic achievement.

Keywords

temperament; development; school outcomes; academic achievement; parental monitoring

Correspondence concerning this article should be addressed to Rongxin Cheng, Department of Psychology, University of California, Davis, One Shields Ave., Davis, CA 95616. rxcheng@ucdavis.edu.

Study materials and R scripts for this project are available on the Open Science Framework (OSF): https://osf.io/jd3vn/?view_only=b5a18949690c40488bae62e7c494e4b2. We are not legally or ethically allowed to publicly post data for this project because the participants in the study have not given informed consent to have their personal data publicly shared, and we do not have IRB approval to post data publicly.

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Success in school can facilitate success in life. Students who do well in school have more educational and occupational opportunities (Irwin et al., 2022; Kautz et al., 2014; McCaul et al., 1992), and those who struggle academically are more likely to experience mental health problems (Björkenstam et al. 2011; Gauffin et al., 2013; Grant et al., 2001) and engage in criminal behavior (Lochner & Moretti, 2004). School outcomes at the transition from adolescence to young adulthood are the most proximal predictors of future educational attainment and labor market earnings; therefore, it is important to understand factors that facilitate or impede academic success at this juncture. Moreover, it is essential to identify factors that contribute to the academic success of minoritized students, including Mexican-origin youth living in the United States, who face a wide range of systemic barriers to pursuing their education, including worse access to educational materials (Hurtado et al., 2020), increased likelihood of living in neighborhoods with socioeconomic distress (Turner et al., 2015; Wildsmith et al., 2016; Altschul, 2012), increased attendance at more poorly funded schools (Crowder & South, 2003; National Equity Atlas, 2018), and more frequent experiences of racism and other forms of discrimination from peers and teachers (Benner & Graham, 2011; Copur-Gencturk, 2020). The impact of these systemic inequities on educational outcomes is particularly salient for youth whose parents have not attended college (Cataldi et al., 2018) and those who are first-generation immigrants (Moosmann et al., 2014; Suárez-Orozco et al., 2010). Despite efforts to reduce achievement gaps, Mexican-origin youth remain at higher risk for school dropout (National Center for Education Statistics [NCES], 2022a), low educational attainment (Irwin et al., 2022; NCES, 2022b), and low-wage employment (U.S. Bureau of Labor Statistics, 2022), compared to White, European-background youth. Notably, these trends stand in stark contrast to the high educational aspirations and expectations of Mexican-origin youth (Lawson et al., 2020).

Despite systemic obstacles they must overcome, many Mexican-origin youth go on to excel academically and in their careers having been influenced by numerous individual, familial, and socio-cultural factors. Understanding individual-level factors that are associated with educational achievement may help identify youth who need more support from parents, teachers, and schools in the current educational system, and these factors may also be leveraged to design more effective community and school-based programs that promote the educational success of all youth. At the individual level, there are multiple factors that influence educational success. Cognitive abilities are not the sole predictor of success in school (Demange et al., 2021; Kautz et al., 2014; Lavrijsen et al., 2022; Noftle & Robins, 2007; Soto et al., 2022); instead, studies have shown that socio-emotional skills (Durlak et al., 2015; Guo et al., 2022; Soto et al., 2022) and the capacity to manage goal-directed behaviors, both of which are related to adolescent temperament, are critical for attaining desirable school outcomes (Robson et al., 2020). Temperament is a particularly important individual factor to consider because it is arguably more malleable than cognitive ability. The present study used data from a longitudinal study of 674 Mexican-origin youth followed from age 10 to 23 to examine how developmental trajectories of temperament (Effortful Control, Negative Emotionality, and Positive Emotionality) from late childhood through adolescence relates to academic performance (e.g., grades, test scores) and educational attainment during late adolescence and young adulthood.

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Temperament and Academic Achievement

Temperament refers to individual differences in reactivity and self-regulation that emerge early in life and provide the foundation for adult personality traits (Rothbart, 2007; Rothbart, 2011). Temperament traits can be organized into three broad domains (Caspi & Shiner, 2006; Rothbart & Bates, 2006): Effortful Control (EC), Negative Emotionality (NEM), and Positive Emotionality (PEM). The EC domain encompasses the capacity to plan and suppress inappropriate impulses (Inhibitory Control), perform an action or pursue goals when there are competing needs and desires (Activation Control), and focus and shift attention when needed (Attention Control). The NEM domain involves negative affect derived from anticipating distress (Fear) and the disruption of ongoing tasks (Frustration), as well as unpleasant affect, lowered mood, and lack of enjoyment in activities (Depressed Mood); in some cases, NEM also includes hostile and aggressive actions (Aggression) and behavioral inhibition to novelty and social challenges (Shyness). Finally, the PEM domain involves pleasure derived from high intensity or novel activities (High Intensity Pleasure/ Surgency; hereafter referred to as Surgency) and a desire for close, warm interpersonal connections (Affiliation). In the present study we consider both domain and facet-level measures of EC and NEM, but we only consider facet-level measures of PEM because Surgency and Affiliation do not form a cohesive domain (Lawson et al., 2021; Snyder et al., 2015).

Although temperament and personality traits are often discussed as separate constructs, Rothbart's temperament domains are theoretically and empirically related to the Big Five personality traits. In particular, EC is linked to Conscientiousness, NEM maps onto Neuroticism, the Surgency facet of PEM shares similarities with Extraversion, and the Affiliation facet of PEM with Agreeableness (DeYoung et al., 2013; Evans & Rothbart, 2007; Putnam et al., 2001; Shiner & Caspi, 2003; Shiner, 2019). Therefore, we organize our review of relevant research using Rothbart's temperament framework but include studies on the Big Five as supplemental evidence. Additionally, we focus on longitudinal studies and only discuss cross-sectional research that might shed light on the mechanisms linking temperament and achievement (see Table 1 for details of longitudinal studies on temperament and academic achievement).

Effortful Control and Academic Achievement

Conceptually, EC may facilitate school success by helping youth to focus and pay attention in class, persist on challenging academic problems, and resist temptations to engage in distracting activities. Indeed, prior research suggests that EC has motivational, emotional, and interpersonal benefits in the school context. Motivationally, high EC promotes goal-directed behaviors such as striving to do well on assignments and starting homework assignments early. Emotionally, high EC is associated with better regulation of negative emotion (Moran et al., 2013; Muris & Ollendick, 2005), which may help students cope with frustrating tasks and reduce disruptive behaviors in school (Atherton et al., 2019; Zhou et al., 2007). Interpersonally, high EC is related to better socio-emotional functioning and more positive student-teacher and peer relationships (Hernández et al., 2017; Valiente et al., 2011; Zhou et al., 2010). Duckworth and Allred (2012) posited that EC-related traits

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like self-regulation play an even more important role during high school and beyond, when workload intensifies and support from teachers decreases.

Consistent with these ideas, numerous longitudinal studies have shown that primary school-aged children (kindergarten to 6th grade) with higher levels of EC are more likely to do well in school (Liew et al., 2008; Valiente et al., 2011; for a review, see Eisenberg et. al., 2011). In adolescence, extant research suggests that EC is prospectively associated with higher standardized test scores (Dindo et al., 2017), higher grades and GPA (Lawson et al., 2021; Valiente et al., 2011, 2013; Zhou et al., 2010), better academic skills (Zorza et al., 2013), and higher educational attainment (Véronneau et al., 2014). Moreover, Conscientiousness is associated with academic achievement both concurrently (Brandt et al., 2020; Israel Brandt, Grund et al., 2022; Mammadov, 2022; Tetzner et al., 2020; for a review, see Spielmann et al., 2022) and longitudinally (Chamorro-Premuzic & Furnham, 2003; Heaven & Ciarrochi, 2008; Israel et al., 2019; Noftle & Robins, 2007; Spengler et al., 2016). Despite this robust evidence for the importance of EC (and the related construct of conscientiousness) for academic success, most prior longitudinal studies included only two waves of data, and none examined whether *changes* in EC are associated with school outcomes.

Negative Emotionality and Academic Achievement

Depressed Mood, Fear, Frustration, and Academic Achievement—The two core facets of NEM –Depressed Mood (depression) and Fear/Frustration (anxiety) – are associated with cognitive, motivational, and interpersonal processes that impede learning and motivation in school (Valiente, Swanson, & Lemery-Chalfant, 2012; Weidman et. al., 2015; Zhou et al., 2010; for a review, see Pekrun & Linnenbink-Garcia, 2012). In particular, youth with higher Depressed Mood tend to be less responsive to incentives and have lower approach motivation (Dickson & Macleod, 2004; Shankman et. al., 2007), which might negatively impact their ability to motivate themselves to work on assignments and study for upcoming exams. Depressed Mood may also contribute to deficits in performance on measures of general cognitive functioning, fluid intelligence, and information processing speed (Rock et al., 2014; Wagner et al., 2015), leading to worse test scores, lower class placements, and fewer experiences of positive reinforcement through higher grades. Decreased motivation may also put youth at increased risk of dropping out of school.

Anxiety can produce task-irrelevant thoughts that have a deleterious effect on concentration, which could impede academic achievement (American Psychiatric Association, 2013; Ma, 1999; Eysenck, 1997; Wine, 1971). Anxious youth (especially those who are more fearful) tend to have stronger avoidance goals (Dickson & Macleod, 2004) and sometimes lower their achievement expectations as a coping mechanism (Abeler et. al., 2011), leading to reduced effort on homework and exams because they are already planning to perform poorly. Similarly, more frustration-prone youth tend to give up quickly when working on challenging tasks, resulting in a lower likelihood of completing complex problems and learning from failure. Therefore, it is likely that youth who are higher in the NEM facets of Fear and Frustration will show worse academic achievement than their less anxious peers. However, it is also possible that some level of anxiety may actually enhance task

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performance among some subgroups (Owens et al., 2014), in which case higher levels of Fear and Frustration could actually enhance, rather than undermine, academic performance.

Consistent with this conceptual background, prior research has shown that both core facets of NEM are linked to poorer school outcomes. A meta-analysis of longitudinal studies found that more internalizing problems, depression, and anxiety, were associated with higher likelihood of school drop-out, but only depression and internalizing problems were associated with worse grades during childhood and adolescence (Riglin et al., 2014). Similarly, more recent meta-analyses found that depression was positively associated with school drop-out (Clayborne et al., 2019) and negatively associated with grades and test scores (Wickersham et al., 2021). Additionally, Neuroticism is consistently associated with poor educational outcomes (for concurrent associations, see Brandt et al., 2020; Mammadov, 2022; Poropat, 2009; Tetzner et al., 2020; for longitudinal associations, see Chamorro-Premuzic & Furnham, 2003; Israel et al., 2019; Spengler et al., 2016). Despite this body of literature, few studies have examined whether *changes* in NEM-related constructs are associated with academic achievement. One longitudinal study found that children with more rapid increases in teacher-reported internalizing symptoms experienced slower increases (or steeper decreases) in teacher-reported academic achievement (reading and math performance) from 4th through 7th grade (Keles et al., 2018). However, two other longitudinal studies did not find any association between changes in internalizing symptoms and academic achievement during early adolescence (Grimm, 2007; Weidman et al., 2015). Given the inconsistent findings, it remains unclear whether *changes* in Depressed Mood, Fear, and/or Frustration prospectively influence academic achievement.

In addition to the core facets of NEM, youth who are prone to Aggression may be less likely to have positive relationships with teachers and peers, which could contribute to a more hostile and less effective learning environment. Indeed, some studies found that aggression predicted lower test scores (Scott et al., 2014) and grades (Busby et al., 2013; Lawson et al., 2021). In contrast, other studies have failed to find significant prospective effects of aggression on adolescents' grades (Caprara et al., 2000).

Finally, higher levels of shyness – a construct sometimes considered part of NEM – might lead to worse academic achievement because shy children experience peer relationship difficulties (Coplan et al., 2008; Tian et al., 2019; Zhang et al., 2017), have trouble in forming positive teacher-child relationships (Arbeau et al., 2010; Rudasill & Rimm-Kaufman, 2009; Rydell et al., 2005), have lower academic engagement (Hughes & Coplan, 2010), and are more sensitive to less positive classroom climates (Gazelle, 2006). The few longitudinal studies suggest that Shyness is negatively associated with academic achievement. Two studies of kindergarteners found that higher levels of teacher-reported (but not parent-reported) shyness predicted lower test scores six months later (Valiente et al., 2010) and higher levels of parent- and childcare provider-reported shyness indirectly predicted lower teacher-reported language and math skills in 1st grade via kids' social problem-solving skills (Walker & Henderson, 2012). Similarly, Zhang et al. (2017) found that higher levels of teacher-reported shyness at age 6 indirectly predicted lower teacher-rated academic performance at age 7, mediated by teacher-rated peer acceptance. Although

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Shyness is associated with worse school outcomes in young children, it is unclear whether this pattern persists into adolescence.

Despite the many theoretical connections between NEM facets and academic achievement, almost no studies have examined whether *changes* in NEM are associated with later achievement. Therefore, more research is needed to better understand the conditions under which *changes* in NEM impact future academic success and failure.

Positive Emotionality and Academic Achievement

Higher levels of PEM may facilitate better academic achievement. In particular, Surgency is associated with approach motivation, which should encourage students to seek out novel challenges (Shiner & Masten, 2002; Shiner et al., 2003). Positive approach-oriented emotions such as joy, hope, and pride are also associated with greater effort and higher academic self-efficacy (Pekrun et al., 2017; Rothbart & Bates, 2006). Potentially offsetting these effects, however, high levels of Surgency may contribute to greater engagement in social activities, which may distract students from concentrating on their schoolwork and lead to lower grades. The other facet of PEM, Affiliation, is associated with cooperativeness (Volk et al., 2011) and perspective-taking skills (Nettle & Liddle, 2008), which may facilitate positive student-teacher relationships and promote achievement in contexts that involve group problem solving and working collaboratively with others.

Longitudinal research on the association between PEM and academic achievement is both scarce and inconsistent. Shiner et al. (2003) found that childhood Surgency was positively correlated with education attainment and work competence in adulthood, although this effect disappeared when controlling for IQ. Using data from the same sample as the present study, Lawson et al. (2021) found that self-reported Affiliation in 7th grade was positively associated with parent-reported school grades in 9th grade, whereas Surgency was not associated with grades. Research on Extraversion and Agreeableness with academic performance showed weak, if not negligible, associations in adolescents (Israel et al., 2019; Spengler et al., 2016).

Changes in Temperament as an Antecedent of Academic Achievement

Numerous longitudinal studies provide strong evidence that higher levels of EC are associated with positive school outcomes (e.g., better grades), whereas higher levels of NEM are associated with negative school outcomes (e.g., school dropout) and dearth of relevant research makes it difficult to understand the prospective effects of PEM on achievement. Notably, there is almost no research examining how trajectories of *change* in any temperament domain or facet are prospectively associated with academic achievement.

Examining temperament trajectories is important because how an adolescent develops (i.e., increasing, decreasing, or not changing) in a particular temperament trait reflects person-environment transactions that play out over time. In particular, changes in temperament may result from ongoing interactions and transactions between the individual and the environmental contexts in which they are interacting. For example, an adolescent who is decreasing in EC while going through middle and high school may experience quite different person-environment transactions over time, including increasingly selecting

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into dysfunctional environments (e.g., associating with deviant and low-achieving peers, selecting less academically challenging courses, disengaging from classes) and evoking and reacting to these environments in dysfunctional ways (e.g., behaving in ways that disrupt positive interactions with teacher, peers, and parents), which in turn could contribute to further declines in EC, creating a vicious cycle over time. In contrast, an adolescent who experience increases in EC during the same time may have selected into, evoked, or reacted to the environmental stimuli in a more adaptive way, creating a virtuous cycle overtime.

These transactions may play out across different times scales, ranging from momentary interpersonal interactions (e.g., being reprimanded by a teacher after acting out in class) to moderate duration (e.g., continued toxic interaction with a particular teacher over the course of a school year) to long-term stable environments (e.g., being part of a stable peer group throughout high school that tends to skip class and engage in substance use and other forms of delinquency) (Hopwood et al., 2022). Changes in temperament can be seen as a descriptive summary of these cumulative temperament-driven transactional processes; for example, a faster increase in EC may characterize a more adaptive way of engaging in person-environment interactions, whereas a faster decrease in EC may suggest a more maladaptive way of engaging. Consequently, changes in temperament may contribute to school outcomes independent of a youth's initial level.

Potential Moderators of the Association between Temperament and Academic Achievement

Mixed findings from past research on temperament and academic outcomes may be due, at least in part, to moderating variables that attenuate the effects for some subgroups of youth. One potential moderator is parental monitoring, or "parenting behaviors involving attention to and tracking of the child's whereabouts, activities, and adaptations" (Dishion & McMahon, 1998, p. 61), which has well-documented positive effects on a child's academic outcomes (Henry et al., 2011; Plunkett & Bamaca-Gomez, 2003; Shumow & Lomax, 2002). Parental monitoring may moderate the association between EC and academic outcomes, such that the association may be weaker for youth raised by parents who engage in high levels of parental monitoring because these parents actively oversee and regulate their child's behavior, making the child's own self-regulatory capacity less consequential. In contrast, children raised by parents who largely refrain from engaging in monitoring may have an even greater need to develop the capacity to regulate their own behavior, strengthening the association between EC and academic outcomes.

With respect to NEM, parental monitoring may buffer youth against the potential negative consequences of high NEM by helping them regulate the disruptive effects of their negative emotions, focus on their academic goals, complete their homework assignments, avoid tardiness and truancy, and reduce engagement in activities outside of school that are detrimental to academic achievement (e.g., doing drugs and associating with deviant peers).

Together, these processes suggest a third potential moderating effect. Specifically, high levels of EC may serve a similar function as parental monitoring, by helping youth regulate their disruptive negative emotions, avoid disengagement and distraction in academic contexts, and maintain positive relationships with classmates and teachers (Eisenberg &

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Fabes, 1992). Research on young children has shown that EC moderates the association between NEM and school achievement, but the direction of effect was not always consistent. Some studies found that the adverse impact of NEM on indicators of academic competence was mitigated by high EC (Diaz et al., 2017; Valiente et al., 2010; Valiente, Swanson, & Eisenberg, 2012). However, it is possible that the moderating effect of EC on NEM and academic outcomes during adolescence is different from that in early childhood. Valiente, Swanson, and Eisenberg (2012) suggest that as children grow older and develop more regulatory skills, teens would be able to regulate high NEM that they were not able to regulate as preschoolers. To date, we know of no studies that have examined such effects in adolescents.

Group Differences in the Association between Temperament and Academic Achievement

Finally, we examined whether the associations between temperament and academic outcomes held for boys versus girls and for youth born in the U.S. versus Mexico. Given the lack of relevant theory and research, these analyses are exploratory and help us establish the generalizability of our findings across gender and nativity.

The Present Study

The present study examined the association between temperament trajectories from late childhood through adolescence and academic outcomes during late adolescence and young adulthood using data from a large, longitudinal study of Mexican-origin youth (N = 674) assessed 12 times from age 10 (5th grade) to age 23 (6 years post-high school)[1]. The present study addressed the following three research questions:

1. Do the level and slope of temperament trajectories (EC, NEM, PEM) from age 10 to 16 predict academic achievement (e.g., high school GPA, achievement test scores, college entrance exam scores, course difficulty in high school, high school graduation, and college attendance[2]) in late adolescence and young adulthood?
    - **a.** Do these associations hold across all facets of each temperament domain?
    - **b.** Do these associations hold across different types of academic outcomes?
2. Does parental monitoring moderate the effects of EC and NEM on academic achievement, and does EC moderate the effect of NEM on academic achievement?
3. Do the findings generalize across gender (i.e., boys vs. girls) and nativity (i.e., youth born in the U.S. vs. Mexico)?

[1]Pre-registration of this study is available on the Open Science Framework (OSF): https://osf.io/jd3vn/?view_only=b5a18949690c40488bae62e7c494e4b2

[2]The pre-registration of this study indicated that we would examine college GPA and college graduation. However, due to the COVID-19 pandemic, we have not completed the most recent wave of data collection, which will provide needed information about cumulative college GPA and college graduation.

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For Research Question 1, we predicted that higher values on the level and slope of EC (i.e., higher initial levels and greater increases) would be associated with higher academic achievement, whereas higher values on the level and slope of NEM would be associated with lower academic achievement. Given the limited previous research, we did not make any specific predictions for Surgency and Affiliation; these are exploratory analyses. For Research Question 1a, we predicted that the direction of the effects would be positive for all three EC facets and negative for all five NEM facets, but the effect size for Depressed Mood will be stronger than for the other facets. For Research Question 1b, we predicted that the effects of EC and NEM would replicate for all indicators of academic success, but the effect sizes will be stronger for more proximal (i.e., assessed in high school) versus distal (e.g., assessed in college) academic outcomes.

For Research Question 2, we predicted that parental monitoring would moderate the effects of EC and NEM on academic achievement. Specifically, we expected that when parental monitoring is high, the effects of EC and NEM on academic achievement would be weak or non-existent, whereas when parental monitoring is low, the beneficial effect of EC and the adverse effect of NEM on academic achievement would be stronger. In addition, we predicted that EC and NEM would have an interactive effect on academic achievement, such that when EC is high, the effect of NEM on academic achievement would be weak or non-existent, whereas when EC is low, NEM would have a stronger adverse effect on academic achievement.

We did not have any specific predictions for Research Question 3. These analyses are exploratory and help us evaluate the generalizability of our findings.

The present study extends previous research in several ways. First, our longitudinal design follows the same participants from late childhood (age 10) through adolescence and into young adulthood (age 23), with 12 total waves of data, allowing us to examine the long-term consequences of temperament for academic outcomes in both high school and college. This extends previous research that has typically included only two waves of data and/or only followed participants over a brief period of time (see Table 1). Second, instead of focusing on a single academic outcome (e.g., test scores), we assessed several different key indicators of academic success, allowing us to evaluate whether the effects of temperament vary across outcomes. Third, we assessed most school outcomes using data obtained from the California Department of Education, allowing us to avoid retrospective and social desirability biases in reports of these outcomes and reduce the risk that any observed associations with temperament are due to shared method variance. This extends previous research that has relied on self-, parent-, and/or teacher-reports of academic performance. Fourth, we used multi-informant assessments of temperament (self and parent-report), which reduces the effect of informant bias, contributes to greater construct validity, and extends past longitudinal research that often relied on only a single informant (Kraemer et al., 2003). Fifth, we used a well-validated measure of temperament to examine both domain- and facet-level effects on academic outcomes, providing a more nuanced understanding of which aspects of each temperament domain are important for academic success. Sixth, we examined these effects in a sample of Mexican-origin youth, which contributes to the generalizability of previous work in this area that has focused on predominantly White,

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European-background children and adolescents. This is especially important given that Latinx youth, and Mexican-origin youth in particular, are impacted by systemic inequities in the American educational system at higher rates than members of racial/ethnic groups, including being more likely to attend under-resourced schools, experience discrimination and acculturative stress, and be raised by parents with a low education levels (Coll & Magnuson, 1997; Irvin et al., 2011; McCoy & Major, 2003; NCES, 2022; Sellers & Shelton, 2003; Suárez-Orozco et al., 2008; The Century Foundation, 2020).

Methods

Participants

This study used data from the California Families Project, an ongoing longitudinal study of Mexican-origin youth and their parents (N = 674).[3] Children were drawn at random from rosters of students from the Sacramento and Woodland, CA school districts. The focal child had to be in 5th grade, of Mexican origin, and living with his or her biological mother in order to participate in the study. Approximately 72.6% of the eligible families agreed to participate in the California Families Project, which was granted approval by the University of California, Davis Institutional Review Board (Protocol # 217484-21). The children (50% female) were assessed annually from fifth grade to 6 years post-high school. The present study uses data from Waves 1–12 (collected in 2006–21), when the children were in fifth grade (M_{age} = 10.9 years, SD = 0.51) to 6 years after high school (M_{age} = 23.2 years, SD = 0.55). Retention rates compared with the original sample are as follows: 84% (Wave 2), 86% (Wave 3), 88% (Wave 4), 90% (Wave 5), 88% (Wave 6), 89% (Wave 7), 89% (Wave 8), 87% (Wave 9), 87% (Wave 10), 80% (Wave 11), and 81% (Wave 12).

Participants were interviewed in their homes in Spanish or English, depending on their preference. Interviewers were all bilingual and most were of Mexican heritage. Sixty-three percent of mothers and 65% of fathers had less than a high school education (median = 9th grade for both mothers and fathers); median total household income was $32,500. With regard to generational status, 83.6% of mothers and 89.4% of fathers were first generation. In the present study, we used data for all available participants (i.e., no exclusions were applied) and reported all analyses conducted to address our research questions.

Attrition Analyses

We compared individuals who did versus did not have school outcome data on the main study variables measured at Wave 1 (i.e., temperament domains and facets, parental monitoring, gender, nativity). We further compared individuals who did versus did not have temperament data at age 12, 14, and 16 on demographic variables (household income, mom education level, dad education level, gender, and nativity). No statistically significant differences were found (see Table S1), with two exceptions. First, numerous temperament domains were associated with having (vs. not having) college entrance exam scores, which

[3]No previous CFP publications examined associations between temperament and any of the school outcomes examined in the present study, except Lawson et al. (2021), which examined associations between self-reported temperament in 7th grade and parent reported school grades in 7th and 9th grade (see Table 1). For a complete list of California Families Project publications, see: Publications - California Families Project. Sample size was determined prior to the CFP data collection.

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simply reflects the fact that participants only take these exams if they intend to go to college. Second, U.S.-born participants were more likely than participants born in Mexico to have data for all school outcomes except college attendance, which probably reflects the fact that participants born in Mexico were more likely to move back to Mexico and consequently no longer attend school in California.

Measures

Temperament—EC, NEM, and PEM were assessed at ages 10, 12, 14, and 16 via self- and mother-reports using the short version of the *Early Adolescent Temperament Questionnaire – Revised* (EATQ; Ellis & Rothbart, 2001). Ratings were made for each item on a 4-point scale ranging from 1 (*not at all true of you/your child*) to 4 (*very true of you/your child*). Given variability in the way the EATQ-R temperament domains are scored (e.g., Lawson et al., 2021; Snyder et al., 2015), we present all findings separately for each individual facet scale, as well as for the broad temperament domains (except Surgency, for reasons noted below). For all domains and facets, we created latent variables using parcels with both self- and mother-reported data.

Effortful control: The EC scale (16 items) has three facets. Inhibitory Control (5 items) assesses the ability to plan and suppress inappropriate impulses. Activation Control (5 items) assesses the ability to perform an action or pursue goals when there are competing desires. Attention Control (6 items) assesses the ability to focus and shift attention when needed.

Negative emotionality: The NE scale (19 items) has three core facets: Fear (6 items), Frustration (7 items), and Depressed Mood (6 items). Fear assesses unpleasant affect derived from anticipating distress. Frustration assesses negative affect related to ongoing tasks being interrupted. Depressed Mood (six items) assesses unpleasant affect, lowered mood, and lack of enjoyment in activities. Two other subscales from the EATQ are sometimes considered part of the NEM domain. Aggression (6 items) assesses hostile reactivity to negative stimuli including person- and object-directed violence. Shyness (4 items) assesses behavioral avoidance of novelty and social challenges. To examine different conceptualizations of NEM, we will run analyses examining a broader conceptualization (including Fear, Frustration, Depressed Mood, Aggression, and Shyness) and narrower conceptualization (including only Fear, Frustration, and Depressed Mood), as recommended by Lawson et al. (2021).

Positive emotionality: The PEM scale has two facets. Surgency (6 or 14 items) assesses pleasure derived from activities involving high intensity or novelty. At age 10, the Surgency scale included the 6 items from the short version of the EATQ-R. Given the low reliability of the 6-item scale at age 10, 8 items were added from the full-length version of the EATQ-R Surgency scale at ages 12, 14, and 16. We modeled the trajectory of Surgency from age 12 to 16 in the current study. Affiliation (5 items) assesses the desire for warmth and closeness with others. Surgency and Affiliation do not form a coherent superordinate factor (Lawson et al., 2021; Snyder et al., 2015); consequently, we did not compute a PEM domain score and instead examined these scales separately.

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School Outcomes—The following school outcomes were obtained from the California Department of Education and the local school districts in Sacramento and Woodland counties: high school GPA, achievement test scores, college entrance exam scores, high school graduation (vs. drop-out), and high school course difficulty (i.e., number of advanced placement [AP] courses taken in high school). When data were missing (e.g., when a child moved to another state or to Mexico during their school-aged years), we used data from self- and parent-reports (if available) to maximize the sample size.

High school GPA: High school GPA was calculated at the end of high school (i.e., end of 12th grade). If the student dropped out of high school, we used the student's cumulative GPA at the year of withdrawal. Following standard practice in California high schools, Advanced Placement (AP) courses were weighted 1 point higher than non-AP courses (e.g., an "A" = 5.0), given the difficulty of getting good grades in these more advanced courses. The resulting variable had 650 cases (M = 2.58, SD = .80, Min = .21, Max = 4.6); 11.2% of the cases are based on self- or parent-reported high school GPA.

Achievement test scores: We created a composite measure of achievement test scores based on the mean of standardized scores on the California Standards Test (CST) from grades 9 to 12 and the California High School Exit Examination (CAHSEE), which was required for all California high school students (N = 587, Min = −2.34, Max = 3.34). All achievement test scores come from school records.

College entrance exam scores: We created a composite measure based on the mean of standardized scores on the Scholastic Assessment Test (SAT) and the American College Testing (ACT) (N = 250, Min = −2.51, Max = 2.59). Overall, 32.8% of the cases were based on self-reported test scores (all were SAT scores). The majority of participants did not take the SAT or the ACT at any point during high school.

High school course difficulty: Course difficulty reflects the number of AP courses that a student took in high school (N = 542, M = 1.22, SD = 2.47, Min = 0, Max = 16). One-hundred and eighty participants (33.2%) took at least one AP course (M in this subsample = 3.66, SD = 3.07); all the data on AP courses come from school records.

High school graduation (vs. drop-out): We created a dichotomous variable indicating whether each participant graduated (coded as 1; n = 541) or dropped out (coded as 0; n = 121) of high school. 12 participants did not report, and their parents did not report, whether they graduated from high school, and no data were available for these participants from the school district or the California Department of Education. Participants who completed high school or its equivalency (GED) were counted as high school graduates; 19% of the cases are based on self- or parent reports of high school graduation. In the current sample, 88% of girls and 76% of boys graduated from high school.

College attendance: We created a dichotomous variable indicating whether each participant attended (coded as 1; n = 350) or did not attend (coded as 0; n = 286) college. 38 participants did not report, and their parents did not report, whether they attended college. Participants attending either four-year colleges/universities or community college were

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classified as attending college (in the current sample, the majority attended community college); participants attending vocational/technical school were not classified as attending college. All cases are based on self- or parent reports of school attendance. In the current sample, 61% of girls and 49% of boys attended college.

Parental Monitoring—Parental monitoring was assessed using a multi-method composite of the Parental Monitoring of Child Scale (PMC; Small & Kerns, 1993) and coding of videotaped mom-child and dad-child conversations. For the PMC scale (available at ages 10, 12, 14, and 16), we used child reports of the mother's and father's monitoring, mother and father reports of their own monitoring, and spousal reports of monitoring (i.e., mother's reports of father's monitoring, and vice versa). Trained coders rated the extent to which the parents monitor and have knowledge of their child's whereabouts after watching parents and children discuss their life together for 20 minutes with the aid of cue cards asking relevant questions (available at ages 10 and 12). Approximately 20–25% of the coding was completed by a second coder to test for reliability. Videos were randomly assigned to coders, with the only constraint that for each parent-child dyad the child was rated by a different coder than the parent. The intraclass correlation between coders was .68 for maternal monitoring and .67 for paternal monitoring. We created a multimethod latent factor of overall "parental monitoring" when the child was 10-, 12-, 14-, and 16-years-old. Each latent factor had three parcel indicators, which were created by averaging across informants (child, parents, spouses) and across methods (PMC and observational data) to minimize the influence of shared method variance (for more details, see Atherton et al., 2020). We then saved the factor scores of the four parental monitoring latent factors (mean-centered) and took the average of the four factor scores for each participant to compute an overall measure of parental monitoring (α = .85).

Nativity—We created a dichotomous nativity variable comparing youth born in Mexico (29%) to youth born in the U.S. (71%).

Data Analysis

Study materials and R scripts for this project are available on the Open Science Framework (OSF): https://osf.io/jd3vn/?view_only=b5a18949690c40488bae62e7c494e4b2. We are not legally or ethically allowed to publicly post data for this project because the participants in the study have not given informed consent to have their personal data publicly shared, and we do not have IRB approval to post data publicly.

Data cleaning and analyses were conducted in R (R Core Team, 2022) via R Studio Version 2022.02.0 and M*plus* Version 8.6 (Muthén & Muthén, 1998–2017) using full information maximum likelihood estimation (FIML) to account for missing data (Allison, 2003; Schafer & Graham, 2002). We used the following R packages to analyze the data and visualize the results: *psych* (Revelle, 2022), *lavaan* (Rosseel, 2012), *ggplot2* (Wickham, 2016), and *semTools* (Jorgensen et al., 2022). All models predicting continuous outcomes with growth curves were conducted in R. All models predicting categorical outcomes with growth curves were conducted in M*plus*. We interpreted good fit as values greater than or equal to .95 for CFI and less than or equal to .06 for RMSEA (Hu & Bentler, 1999). We compared models

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based on a holistic evaluation of multiple criteria: difference in CFI above the threshold of .01, difference in RMSEA was above the threshold of .015, and chi-square difference test (Chen, 2007; Cheung & Rensvold, 2002; Meade et al., 2006), as well as attempting to maximize parsimony.

Research Question 1 was tested using univariate latent growth curve (LGC) models. Longitudinal measurement invariance of temperament domains and facets was tested by comparing the fit indices of configural, weak invariance, partial strong invariance, and strong invariance models. The selected measurement invariance model was retained in the next step where we compared the fit indices of three different LGC models: (1) no growth model, where only the level was estimated; (2) linear growth model, where the slope loadings were fixed to be 0, 2, 4, 6 at age 10, 12, 14, and 16; and (3) latent basis model, where the first and last time points of the slopes were fixed to zero and six, respectively, and the middle time points were freely estimated in order to detect nonlinearities in the growth model. We considered model fit, as well as parsimony, when selecting a LGC model for each domain and facet (for more details about model fitting and comparison, see Lawson, Barrett et al., 2022; Lawson, Bleidorn et al., in press). For the broad and narrow NEM domains that were first examined in the current study, descriptive statistics and reliabilities by wave are shown in Table S2, results of longitudinal measurement invariance analyses are shown in Table S3, and model statistics for the LGC models are shown in Table S4. In the LGC models, one of the school outcomes was regressed on the latent level and slope, separately for each temperament domain and facet. This procedure was repeated for every school outcome. For these analyses, we used an alpha level of .017[4] (two-tailed) to account for multiple comparisons. Additionally, in response to a reviewer's recommendation, we also examined the results using the Benjamini-Hochberg method (1995) to account for multiple comparisons; adjusted-q values are reported in Table S5. Both methods led to the exact same conclusions about statistical significance.

For Research Question 2, we tested the moderating effect of EC using structural equation models with latent variable interactions in M*plus*. In each model, one of the school outcome variables was regressed on one of the three interaction terms between the moderator and predictor (level of the EC × level of NEM, level of EC × slope of NEM, and slope of EC × level of NEM). The moderating effect of parental monitoring was tested by creating interaction terms for parental monitoring, the level, and the slope (parental monitoring × level of the predictor, and parental monitoring × slope of the predictor) to examine whether they had a statistically significant effect on each school outcome.

For Research Question 3, we used multigroup structural equation models to test for gender and nativity differences in the main effects of the temperament trajectories on school outcomes (from Research Question 1). In particular, we compared the fit of models in which the associations between temperament and school outcomes were constrained to be equal across groups to models in which these parameters were freely estimated across groups. If the constrained model did not fit significantly worse than the freely estimated model, then we concluded that the association between temperament and academic success was similar

[4]The adjusted alpha level .017 was derived from 0.05 divided by 3 (3 temperament domains).

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for boys and girls, and for youth born in the U.S. and Mexico. Group differences were tested separately for the level and the slope of the temperament trajectories.

Results

Research Question 1: Associations between Temperament Trajectories and School Outcomes

Zero-order correlations between the temperament trajectories (level and slope) and school outcomes are shown in Table S6 (see Figure 1 for visual depiction of the correlations). Coefficients from LGC models in which the level and slope factors were used to predict each school outcome are shown in Table 2.

Effortful Control—The level of EC and its facets generally predicted higher school achievement (Table 2). Specifically, a one standard deviation (SD) increase in EC and its facets at age 10 predicted a .32 (Inhibitory Control) to .39 (Activation Control) unit increase in high school GPA (standardized coefficients (βs) ranged from .40 to .49 across EC domain and facets, all *ps* < .001). A one SD increase in EC and its facets were associated with a .36 (Activation Control) to .53 (Attention Control) SD increase in achievement test scores (all *ps* < .001). A one SD increase in EC and its facets resulted in .21 (EC) to .37 (Attention Control) SD increase in college entrance exam scores (all *ps* < .01). For high school course difficulty, a one SD increase in EC and its facets were associated with .74 (Activation Control) to .87 (EC) more AP courses taken (βs ranging from .30 to .35, p < .001). A one SD increase in EC and its facets predicted a 52% (Inhibitory Control) to 60% (Effortful Control) increase in the odds of graduating from high school (standardized log odds ranging from .42 to .47, p < .001) and a 39% (Activation Control) to 45% (Effortful Control) increase in the odds of attending college (standardized log odds ranging from .33 to .37, p < .001).

The slope of EC and its facets predicted school outcomes in a few cases after controlling for the baseline level of EC. A faster increase in EC from age 10 to 16 predicted higher high school GPA (β = .28, p < .001), a 25% increase in the odds of graduating from high school (standardized log odds = .22, p = .003), and a 20% increase in the odds of attending college (standardized log odds = .18, p = .009). A more rapid increase in Activation Control predicted higher high school GPA (β = .27, p < .001) but lower college entrance exam scores (β = −.39, p = .011). A faster increase in Attention Control predicted a 42% increase in the odds of graduating from high school (standardized log odds = .35, p = .001). Finally, a steeper slope of Inhibitory Control was associated with a 46% increase in the odds of graduating from high school (standardized log odds = .38, p = .001), and a 31% increase in the odds of attending college (standardized log odds = .27, p = .005).

Negative Emotionality—The level of NEM and its facets generally preceded lower school achievement. In particular, a one SD increase in NEM (broad) at age 10 was associated with .15 decrease in high school GPA (β = −.19, p < .001), which was driven by the Aggression and Depressed Mood facets (β = −.33 and −.19, respectively, p < .001). A one SD increase in NEM (broad) at age 10 was associated with .27 SD decrease in achievement test scores (p < .001), which was driven by the Frustration, Aggression, and

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Depressed Mood facets ($\beta = -.13$, $p = .014$; $\beta = -.29$, $p < .001$; and $\beta = -.24$, $p < .001$, respectively). Therefore, it was not surprising that NEM (narrow) at age 10 also predicted lower achievement test scores ($\beta = -.19$, $p = .001$). For college entrance exam, no statistically significant effect of NEM (and its facets) level was detected except that a one SD increase in Shyness was predicted .22 SD decrease in college entrance exam scores ($p = .006$). For high school course difficulty, a one SD increase in NEM (broad) was associated with .43 decrease in the number of AP courses taken ($\beta = -.18$, $p = .001$), the effect of which was driven by Aggression ($\beta = -.22$, $p < .001$) and Depressed Mood ($\beta = -.17$, $p = .003$). Finally, a one SD increase in NEM (broad) at age 10 predicted 18% decrease in the odds of high school graduation ($\beta = -.20$, $p = .003$), and 23% decrease in the odds of college attendance ($\beta = -.26$, $p < .001$). The effect of NEM (broad) was driven by Frustration (*standardized log odds* = −.19, $p = .002$), Aggression (*standardized log odds* = −.37, $p < .001$ for high school graduation, and −.40, $p < .001$ for college attendance), and Depressed Mood (standardized log odds = −.22, $p = .002$ for high school graduation, and −.22, $p < .001$ for college attendance).

Some effects of the slopes of NEM and its facets were contrary to what we had hypothesized. Controlling for baseline NEM, a more rapid increase in NEM (broad) from age 10 to 16 was associated with higher achievement test scores ($\beta = .24$, $p = .002$), which was driven by the Frustration and Shyness facets ($\beta = .25$, $p = .002$; $\beta = .17$, $p = .005$, respectively). A steeper slope of NEM (broad) was associated with higher college entrance exam scores ($\beta = .34$, $p = .001$), which was driven by the Frustration, Depressed Mood, and Shyness facets ($\beta = .32$, $p = .003$; $\beta = .28$, $p = .014$; $\beta = .26$, $p = .008$, respectively). Therefore, it was not surprising that the slope of NEM (narrow) also had a positive association with college entrance exam scores ($\beta = .26$, $p = .016$). In addition, a faster increase in Shyness predicted greater high school course difficulty ($\beta = .18$, $p = .006$). However, when predicting high school graduation and college attendance, the slope effects were consistent with our hypothesis. Controlling for baseline NEM, a faster increase in NEM (broad) predicted 15% decrease in the odds of attending college (*standardized log odds* = −.16, $p = .016$), which was driven by the Aggression facet (*standardized log odds* = −.22, $p = .003$).

For Surgency and Affiliation, we found no consistent pattern between temperament trajectories and school outcomes. However, we did find a handful of statistically significant positive effects. In particular, a one SD increase in Surgency at age 10 predicted .22 SD increase in college entrance exam scores ($p = .008$). A faster increase in Surgency from age 10 to 16 predicted 23% increase in the odds of graduating from high school (*standardized log odds* = .21, $p = .007$). A one SD increase in Affiliation at age 10 was prospectively associated with a .19 SD increase in high school GPA ($p < .001$), and .37 increase in the number of AP courses taken ($\beta = .15$, $p = .011$). Lastly, a more rapid increase in Affiliation from age 10 to 16 predicted higher achievement test scores ($\beta = .22$, $p = .007$).

Research Question 2: The Moderating Role of Parental Monitoring and EC

Results of all moderation analyses are presented in Table S7 (see Table S8 for a comparison of BIC scores before and after adding the interaction term in the model). We found a

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statistically significant interaction between parental monitoring and the slope of EC in predicting high school GPA ($\beta = -.15$, $p < .001$), achievement test scores ($\beta = -.14$, $p < .001$), and college entrance exam scores ($\beta = -.09$, $p < .001$), controlling for the corresponding main effects (all ps < .001). As shown in Figure 2, when parental monitoring was low (one SD below its mean), more rapid increases in EC from age 10 to 16 strongly predicted higher high school GPA, achievement test scores, and college entrance exam scores, whereas these effects were significantly weaker (but still positive) when parental monitoring was high.

We also found a statistically significant interaction between parental monitoring and the slope of NEM (broad) in predicting high school GPA ($\beta = .16$, $p = .001$), achievement test scores ($\beta = .18$, $p < .001$), and college entrance exam scores ($\beta = .14$, $p < .001$), controlling for the corresponding main effects (all ps < .01). In addition, the interaction between parental monitoring and baseline NEM significantly predicted college entrance exam scores ($\beta = .08$, $p = .004$). As shown in Figure 2, these findings indicate that when parental monitoring was low (one SD below its mean), more rapid increases in NEM from age 10 to 16 predicted lower high school GPA, achievement test scores, and college entrance exam scores, whereas when parental monitoring was high (one SD above its mean), the effect of NEM was negligible. Additionally, when the interaction between parental monitoring and the slope of NEM was added to predict GPA, exam scores, and achievement test scores, the main effect of the slope of NEM turned out to be negative. No statistically significant interactions between parental monitoring and NEM were found for the other school outcomes.

We did not find a statistically significant interaction between the level/slope of EC and the level/slope of NEM (broad) in predicting any of the school outcomes. When predicting high school course difficulty and when predicting academic outcomes with the slope of EC × slope of NEM, the models did not converge.

Research Question 3: Generalizability of Findings Across Gender and Nativity

Results of multigroup comparisons are shown in Table S9 and S10 (for Benjamini-Hochberg adjustment results, see Tables S11 & S12). For all models, constraining the parameters to be equal across groups did not lead to a statistically significant reduction in fit after adjusting the false-discovery rate. For models that required comparing BIC scores, the freely-estimated model tended to have higher BIC scores (worse fit) than the constrained model. Together, these multi-group comparisons indicate that the findings were similar for boys and girls, and for youth born in Mexico and the U.S.

Discussion

The present study used multi-method data from a longitudinal study of 674 Mexican-origin youth to investigate temperamental antecedents of multiple indicators of academic achievement. Specifically, we investigated associations between temperament domain and facet trajectories and six school outcomes (i.e., high school GPA, achievement test scores, college entrance exam scores, high school course difficulty, high school graduation, and college attendance). Additionally, we investigated whether parental monitoring served as a

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moderator of the associations between temperament and school outcomes and whether any associations differed based on gender or nativity.

Effortful Control and Academic Outcomes

Our findings regarding EC at age 10 and subsequent school achievement were consistent with our hypothesis and with prior longitudinal research (Dindo et al., 2017; Valiente et al., 2013; Véronneau et. Al., 2014; and Zorza et al., 2013), such that higher initial levels of EC were associated with better academic outcomes. This effect replicated across all EC facets and academic outcomes, except for college entrance exam scores. The EC-achievement effects were similar in magnitude across the various indicators of school achievement. These findings suggest that the positive association between EC measured in earlier years and school achievement measured years later in predominantly White, European-background samples can be replicated across developmental periods and in a sample of Mexican-origin youth.

Furthermore, we found evidence that more rapid increases in EC and its facets from age 10 to 16 predicted better school achievement after controlling for their corresponding baselines. The slope in Attention Control was particularly important in predicting high school graduation and college attendance, which highlights the focal role of attention-orienting skills in regulatory processes. Indeed, the maturity of Attention Control not only facilitates a focus on goal-relevant stimuli (shift from immediate impulse gratification to attend to long-term goals; Rothbart, Ahadi & Hershey, 1994; Rodriguez & Shoda, 1989), but also promotes socialization such as the development of empathy (shift from one's own perspective to attend to others'; Goodhew & Edwards, 2021; Tusche et al., 2016) and better emotion regulation (shift attention from negative stimuli to positive ones; Kertz et al., 2016; Nozadi et al., 2018; Rothbart & Sheese, 2007). An increase in Attention Control activates a positive feedback loop where the capacity to sustain long-term goals, get along with peers, and regulate emotions codevelop across adolescence, which altogether contribute to higher educational attainment in early adulthood.

In addition, Activation Control played an important role in predicting high school GPA and college entrance exam scores, though its impact on the latter indicator of school achievement was the opposite to our hypothesis. The critical role of Activation Control in predicting high school GPA is not surprising because the scale items used to measure it (e.g., "Being on time for school and appointments", "Put off working on projects" - reverse coded, and "Have hard time finishing things on time" - reverse coded) are directly relevant to course grades. A faster increase in Activation Control suggests increasing ability to overcome academic procrastination, which leads to better grades on assignments, homework, quizzes, and tests, leading to better course grades and a higher overall GPA (Kim & Seo, 2015). Regarding college entrance exam scores, a faster increase in Activation Control predicted lower college entrance exam scores. This result is counterintuitive but can be partially explained in our subsequent discussion about the moderating effect of parental monitoring.

Altogether, the EC findings support past research that the capacity to suppress inappropriate impulses, refrain from distractions in goal pursuit, and focus and shift attention when needed contributes to greater academic achievement.

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Negative Emotionality and Academic Outcomes

Consistent with our hypothesis and prior literature, the broad NEM domain at age 10 predicted worse school outcomes in early adulthood. However, this domain NEM effect should be interpreted cautiously, given the variability in the effects across the various facets of NEM. Aggression and Depressed Mood were most robustly associated with poorer subsequent academic achievement, which is consistent with prior research showing that Aggression (Busby et al., 2013; Scott et al., 2014) and Depressed Mood (Clayborne et al., 2019; Riglin et al., 2014) have more replicable adverse effects on academic outcomes than other NEM facets.

Fear, the experience of negative affect for anticipating distress, showed basically no effect on school achievement in our sample. This finding is not especially surprising given that Fear is an important component of anxiety and prior literature suggests a weak and inconsistent association between anxiety and academic achievement, perhaps because both particularly high and low levels of anxiety may be maladaptive in achievement contexts. To better explore whether there may have been a curvilinear association between the level/slope of Fear and school achievement in our sample, we created quadratic terms of the level/slope of Fear and ran growth curve models in M*plus*. However, we found no significant effect for the quadratic term when predicting any school outcomes. This suggests that there does not seem to be an optimal level of Fear between-persons that maximizes academic performance. It is also possible that people with high levels of Fear do not form a homogenous group because high levels of Fear may impact school achievement through different mechanisms and thereby have different consequences. In particular, the mechanism through which anxiety produces avoidance goals (Dickson & Macleod, 2004), lower academic expectations (Abeler et al., 2011), and thereby lower academic achievement may be only applicable to a subgroup of fearful adolescents.

Another important component of NEM, Frustration, assessed at age 10, was associated with lower achievement test scores, lower high school GPA, and lower likelihood of college attendance, although its effect on the latter two outcomes was not statistically significant. The impact of Frustration on academic outcomes may not have been as consistent because its implications for achievement may vary across individuals. For example, VanSchyndel et al. (2017) found that the effect of Frustration on GPA was moderated by the students' gender and level of impulsivity. Another possibility is that the critical issue is not the experience of frustration itself, but rather the capacity to regulate one's behavior toward achievement despite feeling frustrated (i.e., frustration tolerance; Meindl et al., 2019). Despite these complexities, Frustration still had a more consistent association with academic outcomes than Fear.

Shyness at age 10 was prospectively associated with lower college entrance exam scores. This finding is consistent with research showing that Shyness was associated with lower test scores (Viljaranta et al., 2020), although the association between Shyness and objective measures of achievement is often not as robust as it is for teacher-rated outcomes (Hughes & Coplan, 2010). In the present study, the effect of Shyness at age 10 on other school outcomes was negligible. In summary, of the five facets of NEM, Aggression and Depressed

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Mood had more consistent effects than Fear, Frustration, and Shyness in predicting academic achievement in young adulthood.

When it comes to the change in NEM, contrary to our hypothesis, a more rapid increase in NEM was associated with higher test scores, controlling for baseline NEM. The effect was driven by the effects of Frustration, Depressed Mood, and Shyness, which showed the same associations as the broad NEM domain. These positive slope effects are not consistent with previous longitudinal studies on adolescents, which found null effects of the slope of depression (Grimm, 2007) and internalizing problems (Weidman et al., 2015) on achievement. The findings discussed below regarding the moderating effect of parental monitoring help shed light on this counterintuitive result.

Positive Emotionality and Academic Outcomes

We did not find any consistent impact of Surgency and Affiliation on school outcomes. Instead, the effects of Surgency and Affiliation were specific to certain academic achievement domains. For example, Surgency at age 10 predicted only higher college entrance exam scores. College entrance exams are different from routine tasks such as course assignments and tests in that they are a more intense and novel experience. Students with higher Surgency may be more motivated by the challenge of taking college entrance exams and the potential incentive of receiving a high score, whereas those who are low in Surgency may show less approach motivation in response to the exams and consequently disengage in the test context and receive lower scores. In contrast, the effect of Affiliation may be more prominent for routine tasks (e.g., performing well in courses and attaining higher course grades and taking more AP courses) as students with high Affiliation would seek help from others and cooperate in groups. However, contrary to this interpretation, Surgency was not related to better performance on achievement tests (which, like college entrance exams, happen rarely), whereas Affiliation was. The generally null effects of Surgency on school outcomes is consistent with previous meta-analytic findings indicating that the concurrent association between Surgency and math/reading test scores was not statistically significant (Nasvytienė & Lazdauskas, 2021).

The Moderating Effect of Parental Monitoring

Parental monitoring moderated the effect of both EC and NEM on academic achievement. Consistent with our hypothesis, the effects of EC/NEM were stronger when parental monitoring was low. That is, when parents engaged in little oversight of their child's activities and behaviors, youth who experienced more rapid increases in EC had greater academic achievement during adolescence and in young adulthood, compared to those with flatter slopes or declining levels of EC, and youth who experienced more rapid increases in NEM had worse academic outcomes, compared to those with flatter slopes or declining levels of NEM. In contrast, when youth grew up with parents who closely monitored their behavior and whereabouts, changes in EC and NEM had little or no impact on their academic achievement. Notably, youth who were closely monitored by their parents had higher academic achievement on average than those who were not closely monitored, which is consistent with prior studies (Henry et al., 2011; Plunkett & Bamaca-Gomez, 2003; Shumow & Lomax, 2002). These results suggest that faster increases in EC are particularly

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important for promoting academic performance when parental monitoring is low, and high levels of parental monitoring might buffer youth from the maladaptive effect of increasing levels of NEM. The moderating effect of parental monitoring held for high school GPA, achievement test scores, and college entrance exam scores, but not for course difficulty, high school graduation, and college attendance.

Notably, the compensatory effect of parental monitoring on EC does not imply that adolescents might rely solely on parental monitoring as an external resource and lack motivation to develop higher levels of EC. In contrast, adolescents with higher levels of parental monitoring actually experience a shallower dip in EC during adolescence (see Atherton et al., 2020 for an examination of the development of EC from age 10 to 21 in the present dataset). Instead, the compensatory effect of parental monitoring suggests that close parental supervision could help adolescents with smaller increases in EC achieve their goals and reach a similar level of academic success as those with faster increases in EC. In other words, for adolescents with similar levels of academic achievement, some may benefit more from their own temperament, whereas others may benefit more from parental monitoring.

Analyses of the moderating effects of parental monitoring help us better understand the unexpected slope effects of EC/NEM and their facets. As mentioned above, the slope of EC (and its facets) predicted lower college entrance exam scores. However, when controlling for the main effect of parental monitoring and the interaction between parental monitoring and the EC/NEM slope, the slope of EC (and its facets) predicted higher college entrance exam scores and the slope of NEM predicted lower college entrance exam scores and achievement test scores. These findings, which are partially consistent with our hypotheses for Research Question 1 (and in no cases inconsistent with them), suggest that the unexpected slope effects of EC and NEM we found in response to Research Question 1 might be a statistical artifact. Considering that baseline temperament and the slope of temperament were often significantly associated, and that baseline temperament was often a stronger predictor of academic outcomes than the slope of temperament, it is possible that the effect of slope was distorted when both the level and slope were mutually controlled for each other. Theoretically, experiencing increases in EC and decreases in NEM might initiate adaptive person-environment transactions including attending classes and submitting assignment on time, spending less time with deviant peers, and interacting with teachers, parents, and peers in a positive way. In the long term, these momentary positive transactions contribute to greater academic achievement.

The Moderating Effect of Effortful Control

We did not find evidence supporting the hypothesis that EC served as a protective factor of NEM (i.e., when EC is high, NEM will not matter as much in predicting poor school outcomes compared to when EC is low). To deal with the model non-convergence issue mentioned before, factor scores of the slopes were saved and factor score regression was used to predict the academic outcomes. No statistically significant moderating effect of change in EC was detected. Furthermore, to exclude the possibility that the moderating effect of baseline EC was too weak to be detected, a composite score of EC across all waves was computed as a proxy for the overall level of EC across adolescence. Again, the

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effect of NEM on academic outcomes were not different at different levels of overall EC. This finding is consistent with Gumora and Arsenio's (2002) study where they found that EC did not moderate the effect of academic negative affect on GPA among middle-school students. Future studies should consider examining this effect in different age groups and with different measures of NEM.

Gender and Nativity

In general, the associations between temperament and achievement were similar for boys and girls and for youth born in the U.S. and Mexico. These results suggest that the effects of temperament on school outcomes largely generalize across gender and nativity.

Limitations and Future Directions

The present study has several limitations, which give rise to possible avenues for future research. First, we were not able to fully explain the unexpected slope effects; specifically, the fact that the slope effects of EC and NEM were contrary to our hypotheses when the level effect was controlled, but not when parental monitoring was also controlled. Future studies should replicate these slope effects with different samples, age ranges, and school outcomes to examine whether the unexpected results are statistical artifacts or meaningful associations. Second, future studies should attempt to identify the mechanisms through which temperament leads to school outcomes using mediation analysis or path models. Our interpretation of many of the observed associations involves postulating the role of various cognitive, affective, behavioral, and interpersonal mechanisms, but these potential mediating processes cannot be tested with the data available. Third, given our passive longitudinal design, we are not able to draw causal conclusions about the effects of temperament on academic outcomes, or about the reciprocal effects of academic outcomes on temperament. Other models, including different types of cross-lagged models (Orth et al., 2021), could help disentangle the direction of the effects. For example, Israel, Brandt, Spengler et al. (2022) found cross-lagged effects between the Big Five personality domains and school experiences. Fourth, future studies should examine a broader set of contextualized factors to determine how they influence the strength and direction of the associations between temperament and school outcomes. Previous studies have shown that factors from different levels of Bronfenbrenner's ecological system could impact the development of EC (Atherton et al., 2020; Brandt et al., 2019), but few studies have systematically examined moderators of the association between temperament and academic achievement. Future studies should consider factors from multiple systems such as individual (e.g., educational expectations/aspirations), family (e.g., family SES, sibling deviance, and parental involvement in academic work), peer (e.g., association with deviant peers), school (school SES, school violence, school belongingness, and perceived rejection by teachers), and neighborhood (e.g., neighborhood SES and violence). In particular, it would be interesting to examine whether the effect of temperament on academic outcomes is more pronounced when environmental resources are limited (e.g., when youth are living in impoverished neighborhoods and attending poorly funded schools) compared to environments with more abundant resources. Fifth, given the conceptual and empirical overlap between temperament traits and the Big Five personality domains, it would be

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interesting to examine the extent to which the three temperament domains (and their facets) have incremental validity beyond the Big Five domains in predicting future academic outcomes. Sixth, given the strong theoretical and empirical link between temperament domains and psychopathology, future studies should examine whether temperament domains predict academic achievement over and above the effects of various forms of psychopathology. Seventh, future studies should examine whether the moderating effect of parental monitoring would differ at different developmental stages, and when predicting educational outcomes at different levels of proximity. Because all six indicators of academic achievement in the present study are proximal outcomes, it would be interesting to examine more distal outcomes such as college GPA and college graduation to test whether the moderating effect of parental monitoring will be attenuated when youth are living largely independent of their parent during college, and whether the environmental influences from their current environment (university, peers, etc.) would take over the effect of parental monitoring measured during high school. Notwithstanding these limitations, the present findings provide insight into the specific psychological factors that can help Latinx students thrive despite the harmful inequities they encounter in pursuit of their education.

Conclusion

Understanding factors that facilitate academic success among minoritized students is an important societal concern. The present study examined the association between temperament trajectories across adolescence and academic achievement in early adulthood. Results indicated that higher EC at age 10 and lower NEM at age 10 were associated with better educational outcomes, whereas PEM was generally unrelated to school outcomes. Notably, increases in EC had a stronger impact on academic success when parental monitoring was low, whereas decreases in NEM had a weaker effect on academic achievement when parental monitoring was high. These findings suggest that adolescent temperament is an important factor for promoting academic achievement and setting youth on trajectories towards educational and occupational success in early adulthood.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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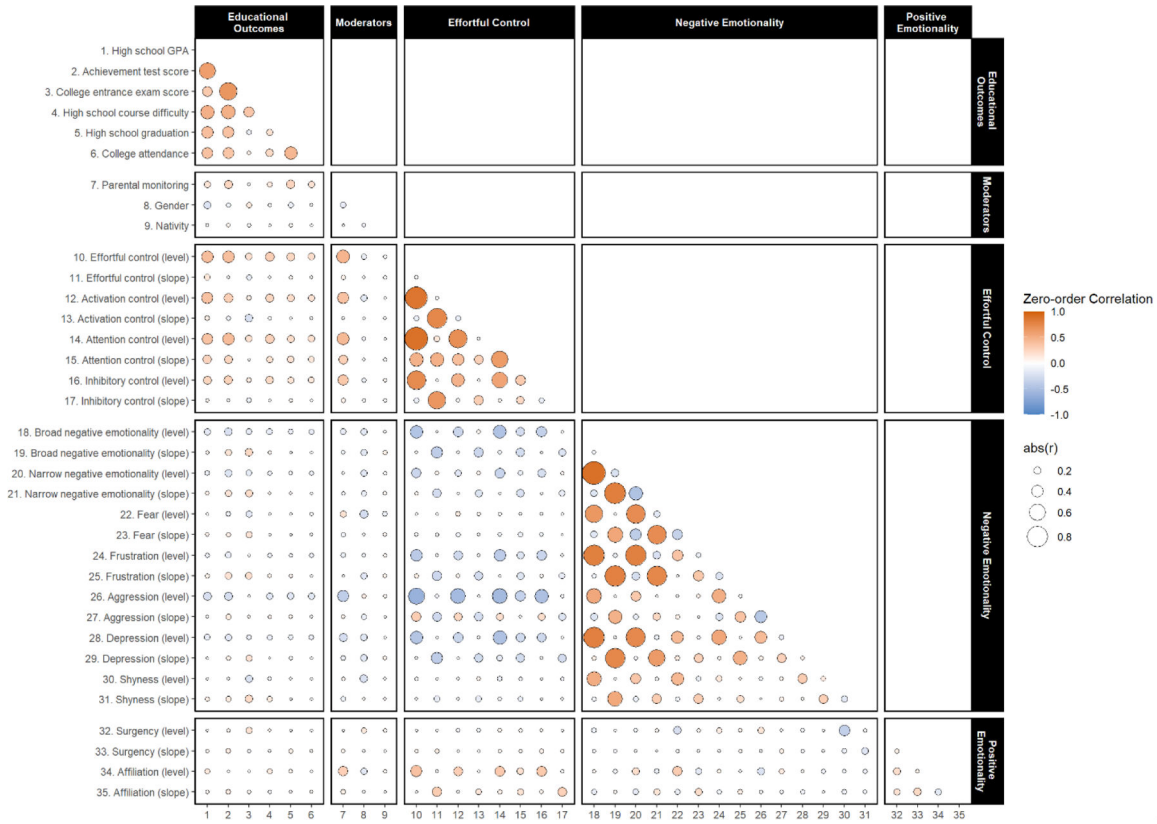

Figure 1.

Zero-Order Correlations between Temperament Trajectories and Academic Outcomes

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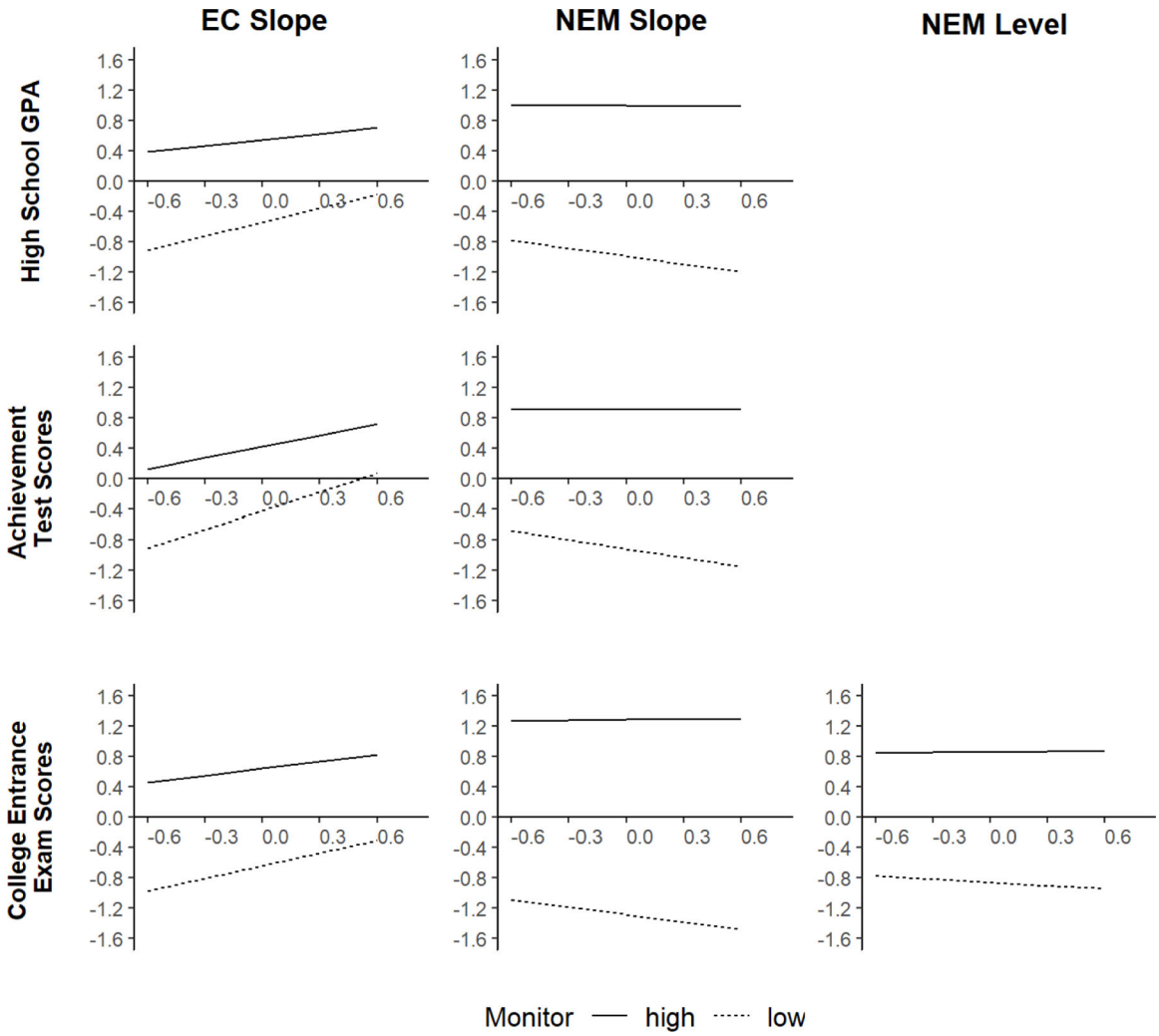

Figure 2. The Moderating Effect of Parental Monitoring on the Association between EC/NEM and Academic Achievement

Note. EC = Effortful Control; NEM = Negative Emotionality; high = One standard deviation above the mean of parental monitoring; low = One standard deviation below the mean of parental monitoring. All variables were standardized.

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Table 1

Summary of Longitudinal Studies Examining Associations between Temperament and Academic Outcomes

Study	Predictor	Assessment method of the predictor	School outcome	Age range[a]	Measurement occasions[b]	Sample size	Country	Summary of findings
Studies of EC								
Dindo et al. (2017)	EC	Behavioral tasks & mother reports	Achievement test scores & mother-reported academic competence	3 to 15	2 times across 12 years	215	U.S.	EC in toddlerhood predicted higher achievement test scores in adolescence.
Lawson et al. (2021)	EC (and facets)	Self-report	Mother-reported school grades	12 to 14	2 times across 2 years	674	U.S.	EC predicted higher school grades in 9th grade
Valiente et al. (2011)	EC	Parent and teacher reports	Self- and teacher-reported grades	7 to 13	2 times across 6 years	214	U.S.	EC at age 7 predicted higher academic achievement at age 13, mediated by child's social functioning.
Valiente et al. (2013)	EC	Parent and teacher reports	Parent- and teacher-reported grades	12 to 14	2 times across 2 years	125	U.S.	EC at age 12 predicted higher academic achievement at age 14
Véronneau et al. (2014)	EC	Self-, parent- and teacher reports	Education attainment	16 to 23	2 times across 6 years	997	U.S.	EC at age 16 predicted higher GPA at age 23
Zorza et al. (2013)	EC	Self-report	Grades and teacher-reported academic skills	12 to 14	2 times across 6 months	359	Spain	EC at age 12 to 14 predicted higher academic achievement 6 months later
Zhou et al. (2010)	EC	Parent- and teacher-reports	GPA in school records	7 to 13	2 times across 4 years	382	China	EC in childhood predicted higher GPA in adolescence, adjusting for GPA in childhood.
Studies of NEM								
Clayborne et al. (2019)	Depression	-	School dropout, tertiary training, post-secondary education	Adolescence to adulthood	Meta-analysis	2905 to 21004, depending on the outcome variables	-	Adolescent depression predicted failure to complete secondary school and lower odds of being in tertiary training and post-secondary education in adulthood.
Riglin et al. (2014)	Depression, Anxiety, Internalizing problems	-	School dropout, grades	Childhood to adulthood	Meta-analysis	24353	-	Depression, anxiety, and internalizing problems were prospectively associated with higher likelihood of school dropout. Depression and internalizing problems also predicted lower school grades.
Wickersham et al. (2021)	Depression	-	Grades, test scores	Childhood to adulthood	Meta-analysis	36528	-	Depression was prospectively associated with lower grades or test scores.
Kwon et al. (2018)	NEM	Teacher-report	Achievement test scores on reading	9 to 13	2 times across 14 months	199	U.S.	No statistically significant association was found between NEM (teacher-report)

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Study	Predictor	Assessment method of the predictor	School outcome	Age range[a]	Measurement occasions[b]	Sample size	Country	Summary of findings
								in late childhood and early adolescence and test scores a year later.
Keles et al. (2018)	Internalizing symptoms	Teacher-report	Teacher-reported academic performance	9 to 12	4 times across 4 years	4458	Norway	Bivariate latent growth curve modeling showed that the slope of internalizing symptoms and the slope of academic performance were negatively correlated. The level of internalizing symptoms did not predict the slope of academic performance.
Lawson et al. (2021)	NEM (and facets)	Self-report	Mother-reported school grades	12 to 14	2 times across 2 years	674	U.S.	NEM in 7th grade did not predict school grades in 9th grade; None of the facets of NEM except for Aggression in the 7th grade predicted school grades in 9th grade.
Weidman et al. (2015)	Internalizing symptoms	Self-report	GPA in school records	11 to 16	5 times across 5 years	130	U.S.	Higher internalizing symptoms at the beginning of the school year predicted lower GPA at the end of the school year in 7th, 9th, and 10th (but not 8th) grade. No statistically significant association was found between the slopes.
Grimm (2007)	Depression	Parental reports	Achievement test scores on vocabulary	8 to 14	7 times across 7 years	7078	U.S.	No statistically significant association was found for the slope of depression and the slope of test scores.
Busby et al. (2013)	Aggression	Teacher-report	Teacher-reported grades, reading achievement, academic readiness	11 to 15	2 times across 1 year	491	U.S.	Aggression in 7th grade predicted lower academic functioning in 8th grade.
Caprara et al. (2000)	Aggressive behaviors	Self-, teacher-, and peer-reports	Teacher-reported course grades	8 to 13	2 times across 5 years	294	Italy	No statistically significant association was found between aggression in 3rd grade and academic achievement in 8th grade.
Scott et al. (2014)	Aggression	Self-report	Achievement test scores in school records	8 to 15	2 times across 6 months	191	U.S.	Aggression predicted lower test scores 6 months later.
Zhou et al. (2010)	Anger, Frustration	Parent- and teacher-reports	GPA in school records	7 to 13	2 times across 4 years	382	China	Higher anger/frustration (teacher-report) in childhood predicted lower GPA in adolescence, but the effect was no longer statistically significant after controlling for GPA in childhood.
Studies of PEM								
Shiner et al. (2003)	Surgency, Agreeableness	Self-, parent-, and teacher-reports	Self- and parent-reports and clinical ratings on education attainment	8 to 32	2 times across 20 years	205	U.S.	Surgency and Agreeableness in late childhood predicted higher education attainment in adulthood.

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Study	Predictor	Assessment method of the predictor	School outcome	Age range[a]	Measurement occasions[b]	Sample size	Country	Summary of findings
Lawson et al. (2021)	Surgency, Affiliation	Self-report	Mother-reported school grades	12 to 14	2 times across 2 years	674	U.S.	Affiliation in 7th grade predicted higher school grades in 9th grade; Surgency in 7th grade did not predict school grades in 9th grade

Note. EC = Effortful Control; NEM = Negative Emotionality; PEM = Positive Emotionality.

[a]Studies were included only if the age range of their sample overlapped with the age range of the present sample (10 to 23 years). For studies that only reported grade in school, we converted the grade range to the approximate age range (e.g., 5th to 7th grade was converted to 10–12 years).

[b]For measurement occasions, we only counted the number of occasions when temperament and/or school outcomes were assessed. We did not count the occasions when other variables were assessed.

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Table 2

Associations Between Temperament Trajectories and School Achievement

	High School GPA	Achievement Test Scores	College Entrance Exam Scores	High School Course Difficulty	High School Graduation	College Attendance
Effortful Control						
Level	**.48 [.38, .59] (*p*< .001)**	**.48 [.37, .58] (*p*< .001)**	**.21 [.02, .40] (*p*= .005)**	**.35 [.24, .47] (*p*< .001)**	**.47 [.31, .63] (*p*< .001)**	**.37 [.23, .51] (*p*< .001)**
Slope	**.28 [.13, .44] (*p*< .001)**	.11 [−.06, .29] (*p* = .069)	−.18 [−.44, .08] (*p* = .096)	.09 [−.09, .26] (*p* = .203)	**.22 [.03, .40] (*p*= .003)**	**.18 [.00, .36] (*p*= .009)**
Activation Control						
Level	**.49 [.37, .61] (*p*< .001)**	**.36 [.24, .48] (*p*< .001)**	.10 [−.10, .31] (*p* = .216)	**.30 [.17, .43] (*p*< .001)**	**.45 [.29, .62] (*p*< .001)**	**.33 [.18, .49] (*p*< .001)**
Slope	**.27 [.08, .46] (*p*< .001)**	−.09 [−.31, .13] (*p* = .319)	**−.39 [−.70, .08] (*p*= .011)**	−.03 [−.23, .18] (*p* = .746)	.07 [−.14, .29] (*p* = .377)	.13 [−.07, .33] (*p* = .101)
Attention Control						
Level	**.43 [.28, .59] (*p*< .001)**	**.53 [.38, .68] (*p*< .001)**	**.37 [.12, .61] (*p*< .001)**	**.33 [.19, .52] (*p*< .001)**	**.43 [.25, .62] (*p*< .001)**	**.37 [.21, .53] (*p*< .001)**
Slope	.12 [−.21, .45] (*p* = .316)	−.02 [−.36, .32] (*p* = .886)	−.41 [−.99, .16] (*p* = .146)	−.01 [−.36, .34] (*p* = .934)	**.35 [.07, .64] (*p*= .001)**	.16 [−.09, .40] (*p* = .100)
Inhibitory Control						
Level	**.40 [.24, .56] (*p*< .001)**	**.41 [.25, .57] (*p*< .001)**	.15 [−.08, .39] (*p* = .114)	**.32 [.16, .47] (*p*< .001)**	**.42 [.22, .62] (*p*< .001)**	**.36 [.18, .54] (*p*< .001)**
Slope	.31 [.03, .58] (*p* = .030)	.27 [−.02, .56] (*p* = .049)	−.20 [−.62, .23] (*p* = .286)	.05 [−.37, .47] (*p* = .76)	**.38 [.10, .67] (*p*= .001)**	**.27 [.02, .52] (*p*= .005)**
NEM (broad)						
Level	**−.19 [−.31, −.07] (*p*< .001)**	**−.27 [−.39, −.14] (*p*< .001)**	−.18 [−.37, .02] (*p* = .023)	**−.18 [−.31, −.05] (*p*= .001)**	**−.20 [−.36, −.03] (*p*= .003)**	**−.26 [−.41, −.11] (*p*< .001)**
Slope	.08 [−.08, .23] (*p* = .205)	**.24 [.07, .40] (*p*= .002)**	**.34 [.11, .58] (*p*= .001)**	.09 [−.08, .26] (*p* = .191)	−.01 [−.21, .20] (*p* = .948)	**−.16 [−.34, .01] (*p*= .016)**
NEM (narrow)						
Level	−.11 [−.24, .19] (*p* = .030)	**−.19 [−.33, −.06] (*p*= .001)**	−.09 [−.31, .12] (*p* = .268)	−.11 [−.25, .04] (*p* = .062)	−.13 [−.30, .05] (*p* = .063)	**−.19 [−.34, −.04] (*p*= .001)**
Slope	.05 [−.11, .21] (*p* = .439)	.14 [−.04, .32] (*p* = .063)	**.26 [.01, .51] (*p*= .016)**	.00 [−.17, .18] (*p* = .968)	.01 [−.22, .23] (*p* = .954)	−.14 [−.32, .05] (*p* = .059)

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		High School GPA	Achievement Test Scores	College Entrance Exam Scores	High School Course Difficulty	High School Graduation	College Attendance
Fear							
	Level	.04 [−.11, .19] (p = .523)	−.11 [−.27, .04] (p = .072)	−.19 [−.41, .05] (p = .058)	−.00 [−.16, .15] (p = .962)	−.01 [−.22, .20] (p = .897)	−.09 [−.29, .10] (p = .205)
	Slope	.16 [−.07, .38] (p = .102)	.15 [−.12, .42] (p = .215)	.28 [−.13, .69] (p = .148)	−.03 [−.27, .22] (p = .776)	−.04 [−.33, .25] (p = .743)	−.16 [−.41, .09] (p = .099)
Frustration							
	Level	−.11 [−.24, .02] (p = .033)	**−.13** **[−.27, .00]** **(p= .014)**	.04 [−.17, .25] (p = .623)	−.10 [−.24, .04] (p = .060)	−.09 [−.26, .09] (p = .186)	**−.19** **[−.35, −.03]** **(p= .002)**
	Slope	.10 [−.07, .27] (p = .131)	**.25** **[.07, .43]** **(p= .002)**	**.32** **[.07, .58]** **(p= .003)**	.10 [−.08, .29] (p = .167)	.14 [−.11, .39] (p = .149)	−.03 [−.23, .17] (p = .707)
Aggression[a]							
	Level	**−.33** **[−.46, −.20]** **(p< .001)**	**−.29** **[−.42, −.16]** **(p< .001)**	.06 [−.20, .31] (p = .576)	**−.22** **[−.36, −.08]** **(p< .001)**	**−.37** **[−.54, −.20]** **(p< .001)**	**−.40** **[−.56, −.24]** **(p< .001)**
	Slope	−.15 [−.33, .04] (p = .046)	.04 [−.17, .25] (p = .655)	.12 [−.28, .52] (p = .441)	−.03 [−.25, .19] (p = .756)	−.11 [−.31, .09] (p = .438)	**−.22** **[−.40, −.03]** **(p= .003)**
Depressed Mood							
	Level	**−.19** **[−.32, .06]** **(p< .001)**	**−.24** **[−.38, −.10]** **(p< .001)**	−.15 [−.37, .06] (p = .076)	**−.17** **[−.32, −.03]** **(p= .003)**	**−.22** **[−.40, −.03]** **(p= .002)**	**−.22** **[−.38, −.06]** **(p< .001)**
	Slope	.00 [−.17, .18] (p = .947)	.16 [−.03, .35] (p = .050)	**.28** **[.02, .54]** **(p= .014)**	.02 [−.17, .20] (p = .821)	−.07 [−.27, .14] (p = .419)	−.14 [−.32, .05] (p = .056)
Shyness							
	Level	−.01 [−.13, .12] (p = .899)	−.02 [−.15, .11] (p = .704)	**−.22** **[−.41, −.02]** **(p= .006)**	−.06 [−.19, .08] (p = .262)	.01 [−.16, .18] (p = .924)	−.09 [−.25, .07] (p = .137)
	Slope	.13 [−.01, .27] (p = .024)	**.17** **[.02, .32]** **(p= .005)**	**.26** **[.03, .48]** **(p= .008)**	**.18** **[.02, .34]** **(p= .006)**	.01 [−.17, .19] (p = .872)	−.14 [−.32, .04] (p = .042)
Surgency							
	Level	−.01 [−.14, .12] (p = .834)	.04 [−.10, .19] (p = .447)	**.22** **[.02, .42]** **(p= .008)**	.05 [−.09, .19] (p = .385)	−.01 [−.21, .19] (p = .936)	.02 [−.14, .19] (p = .753)
	Slope	.08 [−.11, .27] (p = .301)	.19 [−.02, .40] (p = .061)	−.10 [−.39, .20] (p = .445)	.04 [−.16, .24] (p = .631)	**.21** **[−.09, .50]** **(p= .007)**	.09 [−.15, .32] (p = .346)
Affiliation							
	Level	**.19** **[.05, .32]** **(p< .001)**	.03 [−.12, .19] (p = .584)	−.01 [−.23, .21] (p = .912)	**.15** **[.01, .30]** **(p= .011)**	.13 [−.06, .33] (p = .083)	.08 [−.09, .26] (p = .221)
	Slope	.02 [−.06, .30] (p = .077)	**.22** **[.03, .40]** **(p= .007)**	.09 [−.17, .35] (p = .378)	−.07 [−.26, .12] (p = .363)	.19 [−.03, .42] (p = .029)	.12 [−.08, .31] (p = .128)

Note. Values in the table are coefficients from growth curve models in which the latent level and slope factors were used to predict each school outcome, separately for each temperament domain and facet. All values are standardized regression coefficients, except for the values for high school graduation and college attendance, which are standardized log odds. Values in bold were significant at $p < .016$ (based on the Benjamini-Hochberg method); values in brackets are 99% confidence intervals, and therefore depict a stricter restriction on the false discovery rate than the .016 p-level used in the present study.

[a]In the current study, we fit a linear growth curve model to Aggression to examine the slope effect. The normative growth model for Aggression was the no-growth curve model (see Lawson, Bleidorn et al., (in press)).

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